

Benchmarking Decoders for Directional Intent Control of a Mobile Gait Trainer Using a Dry-Electrode EEG-Based BCI

Master's Thesis Defense

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Outline

1. Introduction
 2. Building the experiment
 3. Designing the benchmark
 4. Results & Discussion
 5. Conclusion
- Methods

The problem

Worldwide

12 million

new strokes every year

94 million

living with the effects of stroke

15.4 million

living with spinal cord injury

WSO Global Stroke Fact Sheet 2025
WHO Spinal Cord Injury fact sheet 2024

- Rehabilitation: restore mobility
- Frequent intention-driven practice
- Therapist can't apply ideal intensity

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u^b

The solution?

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u^b Two devices, one idea

Both available thanks to the neuro-rehaLab

Introduction

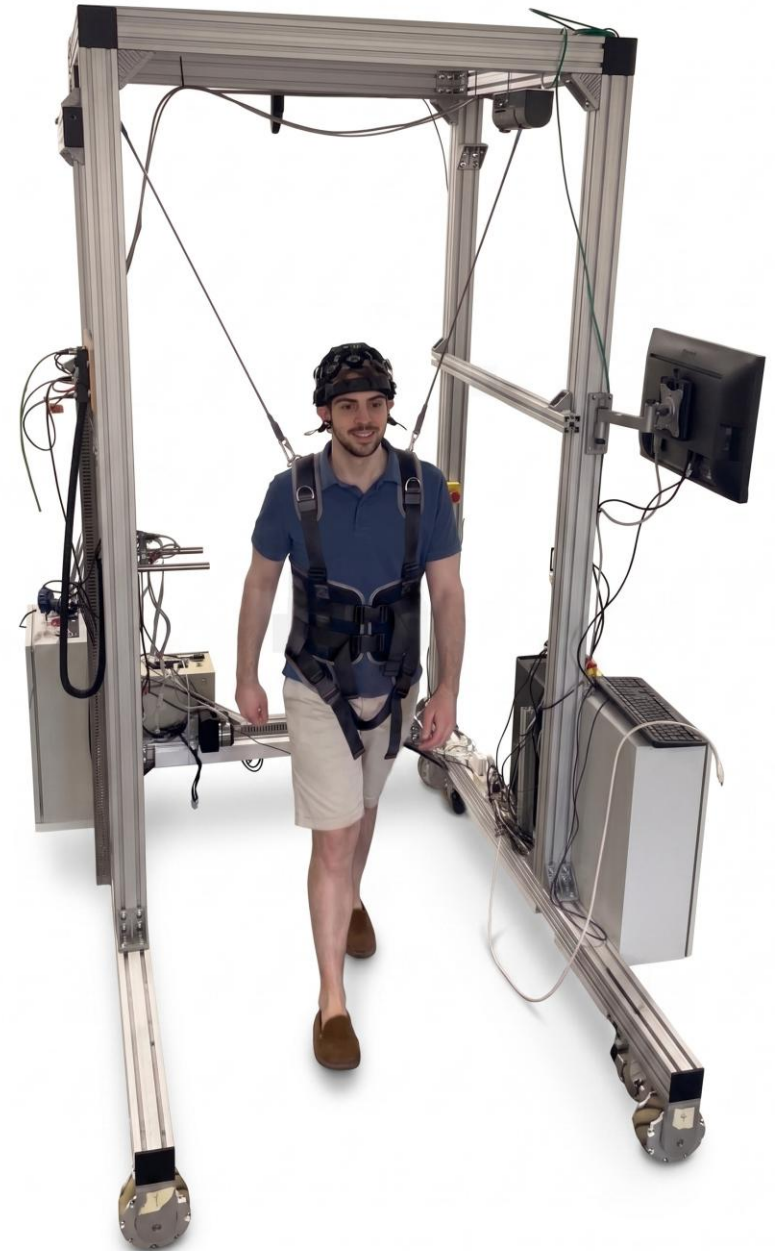
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- DSI-24 dry EEG wireless headset
- Gravity-supported mobile gait trainer
- The idea: let intention steer it



Off-the-Shelf vs Custom Decoders

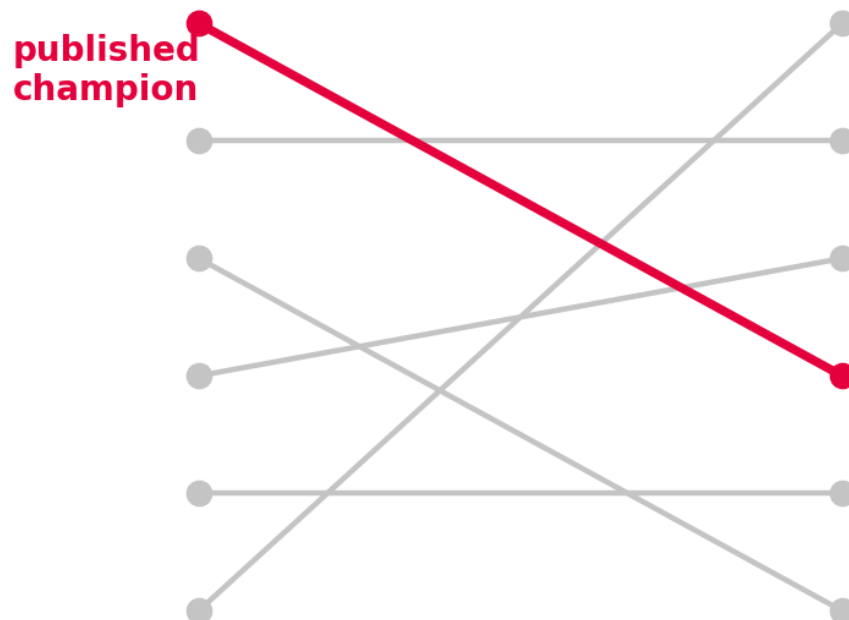
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The field is crowded, but untested

- Focus on what actually works for mobile BCI

No architecture is dominant

- Brigato et al. finds no champion exists yet

Leaderboards do not transfer

- Rankings change with data, task and protocol

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Aim & goal

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1

Restore independent **ambulation** to an impaired **patient** using a supportive rehabilitation **device**

2

Find an **architecture** that can **decode** intended **direction** of movement from brain activity using **EEG**

u^b Five research questions

RQ0 Is direction really decoded, or just movement?

RQ1 Do EEG foundation models beat training from scratch?

RQ2 Does the pretraining corpus matter at all?

RQ3 Who wins when calibration data is scarce?

RQ4 Can the winner drive the robot safely, live?

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u^b Building the experiment

Procedure, software, dataset

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The task

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- **Five cued directional commands**
Record 15s clips
- **Movement activation, not motor imagery**
No public dataset
- **Rapid model decisions**
1.5s from intention to movement

The pipeline

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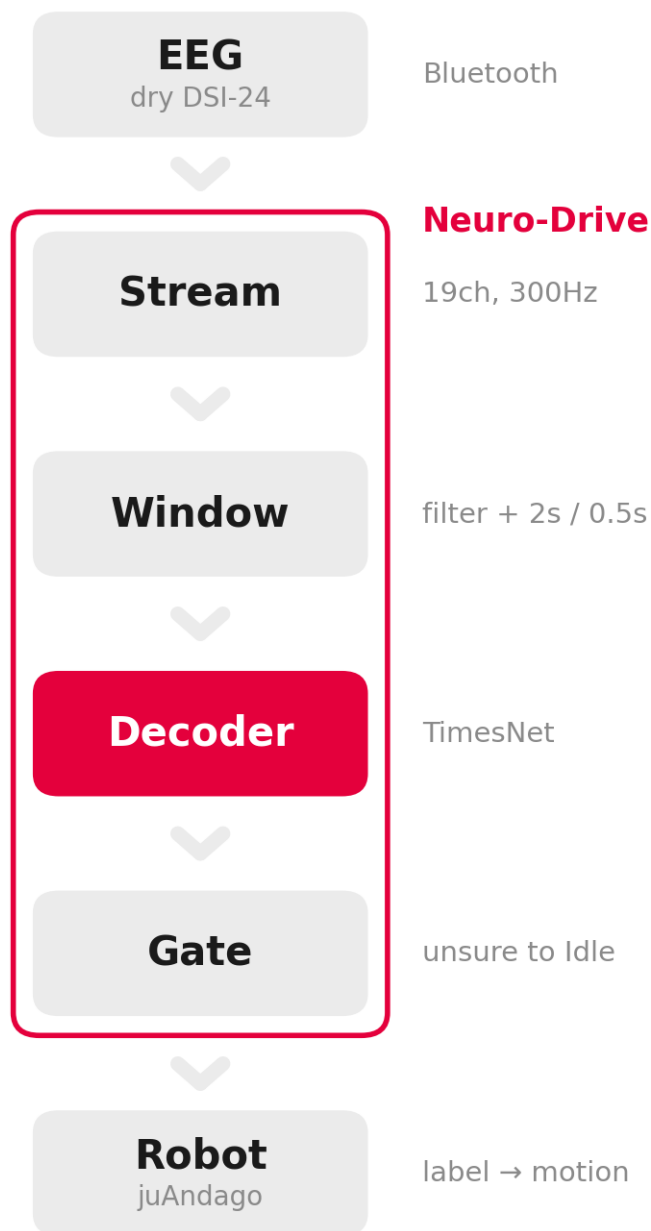
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Bluetooth stream, shared reader

One terminal menu

- Recording free or cued
- Live dashboard, built in PyQt6
- Neuro-Drive: filters, window, inference



The data

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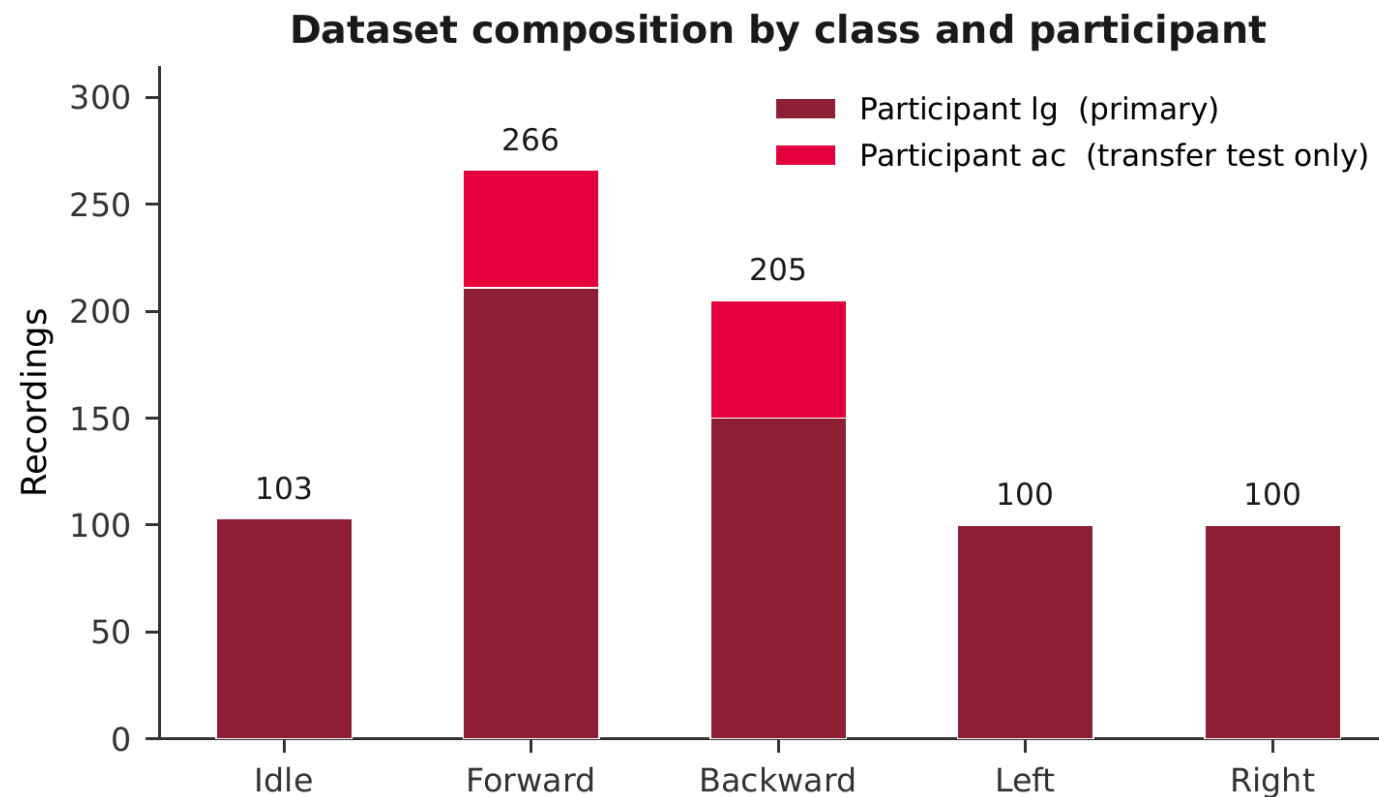
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- **One move per recording**
Different task types

- **Benchmark pool 500**
100 per class

- **Second participant**
Held out Protocol C

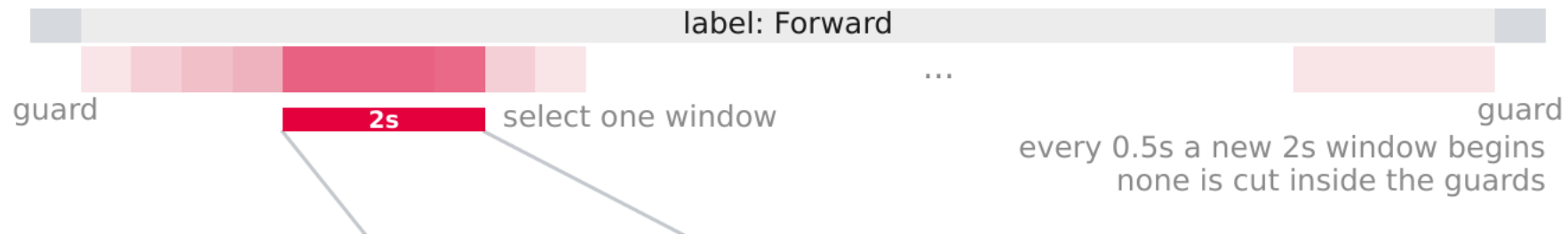


774 recordings of 15 seconds, 20 task styles

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From windowed signals to model inference

1. recording



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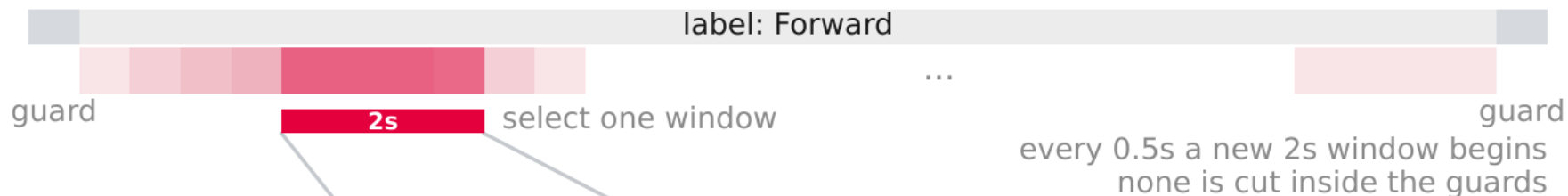
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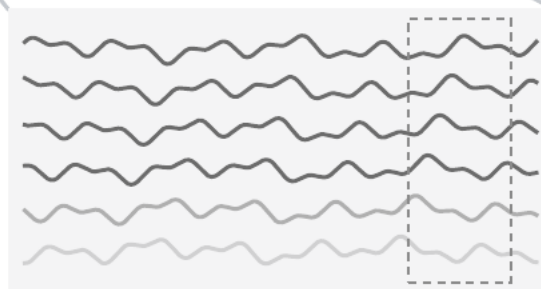
From windowed signals to model inference

1. recording

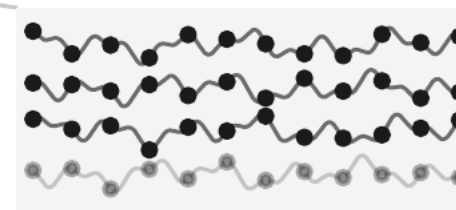


2. window

the target label comes from the recording the window was cut from

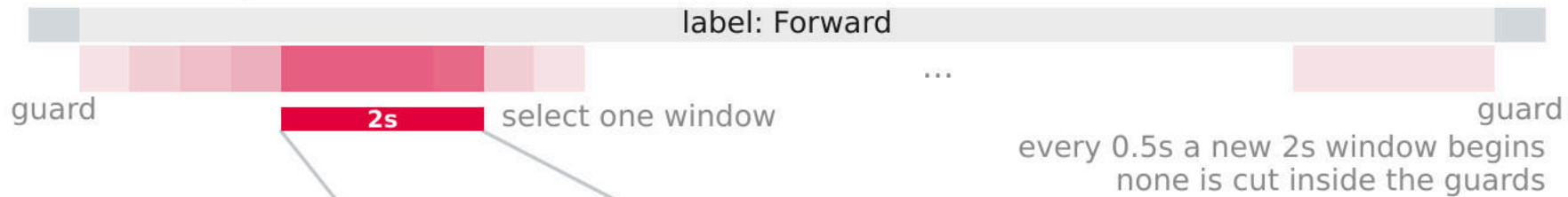


the window's 19 channels



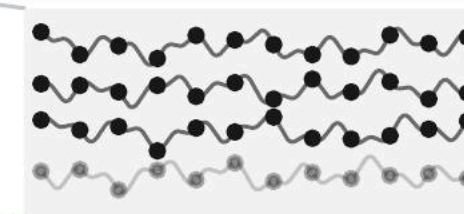
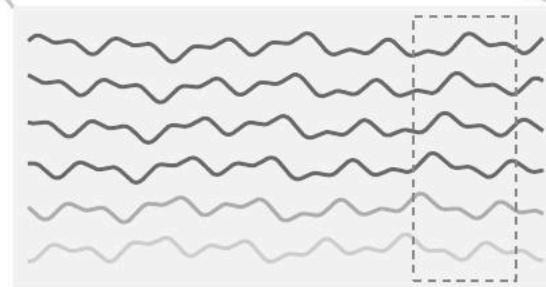
samples at discrete time steps
600 per channel

1. recording



2. window

the target label comes from
the recording the window
was cut from



✓ decoder input
 $2s \rightarrow 19 \times 600$

3. prediction



in training gradient descent updates the weights on every window
right guesses are made more confident, wrong ones corrected harder

u^b

Designing the benchmark

Nine decoders, one classifier, three protocols

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Nine decoders, four families

Each in its strongest form, everything else held equal

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1

Classical baseline

FBCSP+LDA

Per band, the electrode mix that separates classes

2

Purpose-built

EEGNet

Learns frequencies and electrodes

3

General time series, trained from scratch

TimesNet

Compares stacks of repeating rhythms

iTransformer

Attention across electrodes

S-Mamba

Running state over electrodes

Nine decoders, four families

Each in its strongest form, everything else held equal

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4a EEG foundation models, pretrained

Neuro-GPT Predict next chunk from those before

CBraMod Splits attention between electrodes and time

LaBraM Signal patches turned into discrete codes

4b Non-EEG control

Chronos Electrodes quantised into tokens read by LM

Augmentation searched per model, from 18 transforms

Classification head

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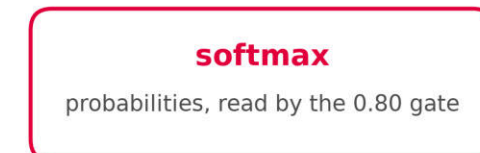
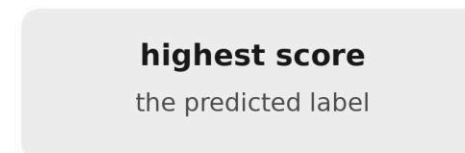
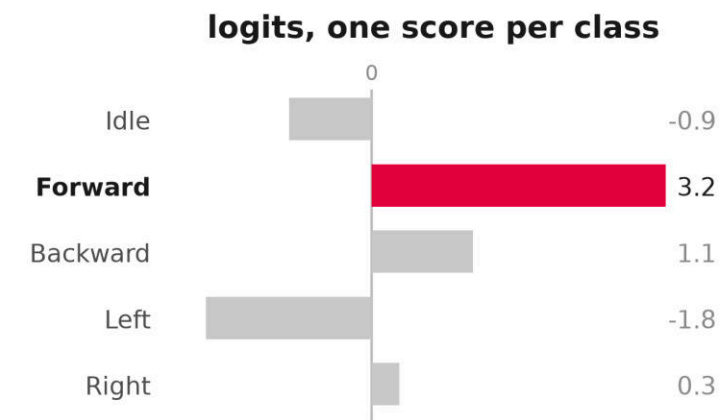
Conclusion

One scoring layer for each backbones

- Input width changes
- Always trained

FBCSP is the exception

- No head, LDA classifies features



Three protocols, three questions

each block is one recording

80%

10%

10%

Protocol A
within-subject



one class shown

Protocol B
cross-recording



one fold of five

Protocol C
cross-person



reference



participant lg

forward + backward only, one split, no folds

participant ac

train validation held out for test

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Results & Discussion

Outcomes of the final benchmark

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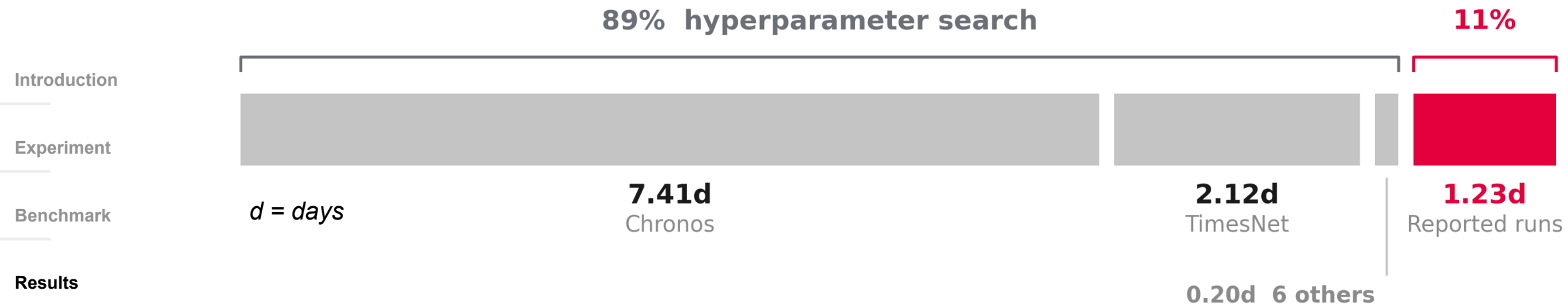
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First complete run on protocol A



- The fair comparison cost 9x more than the results
- The control was the single largest cost

Single harness failure point

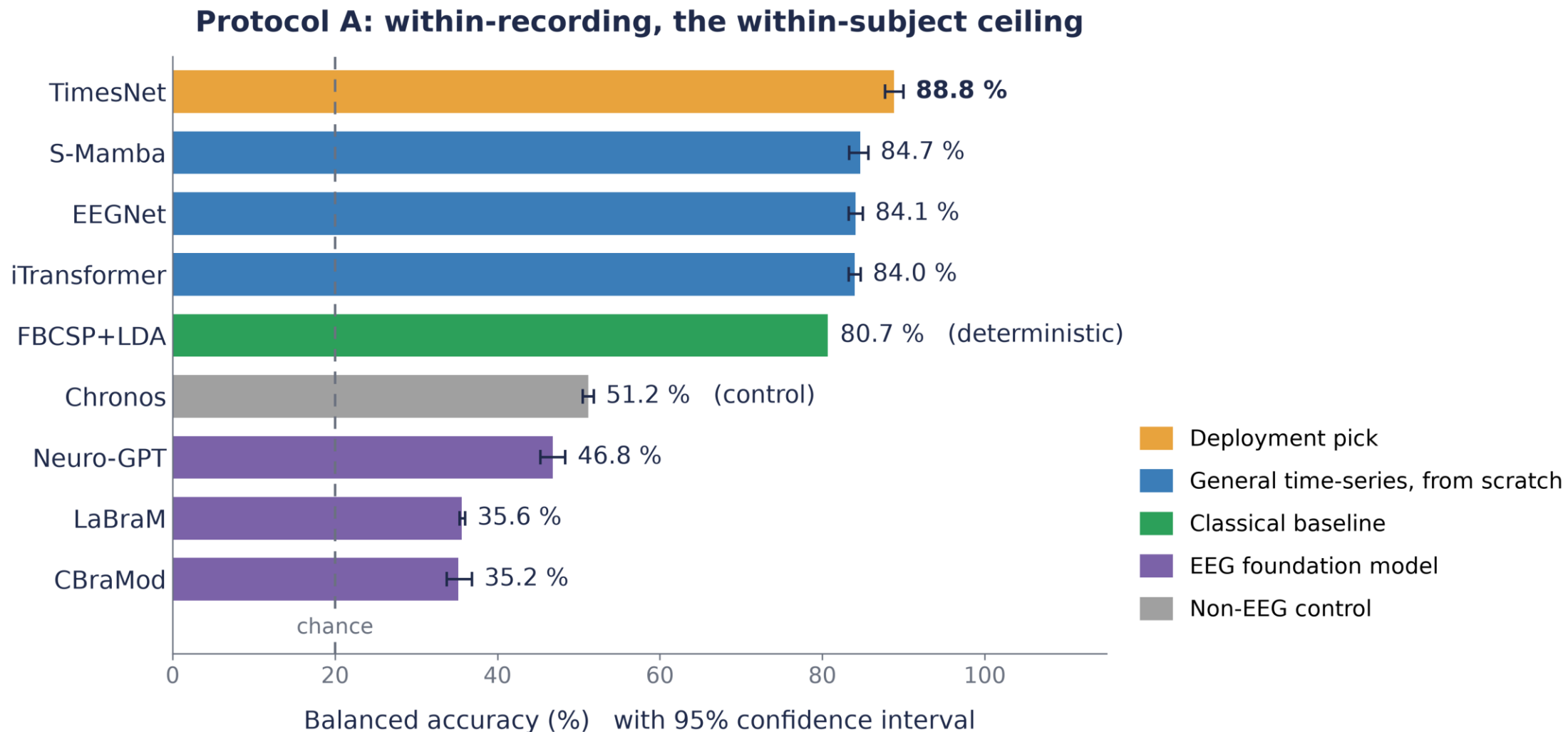
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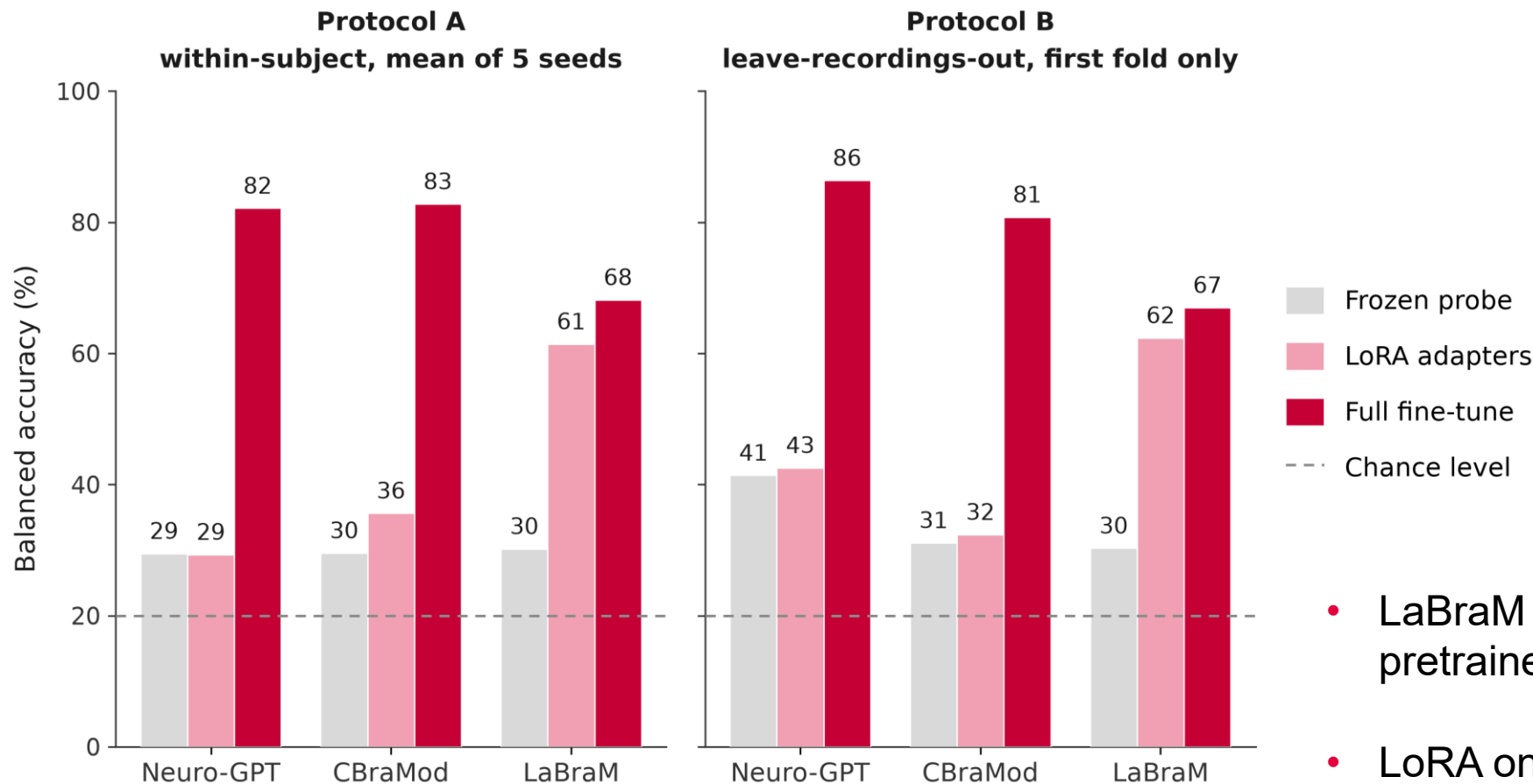
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Pretrained, on their own terms

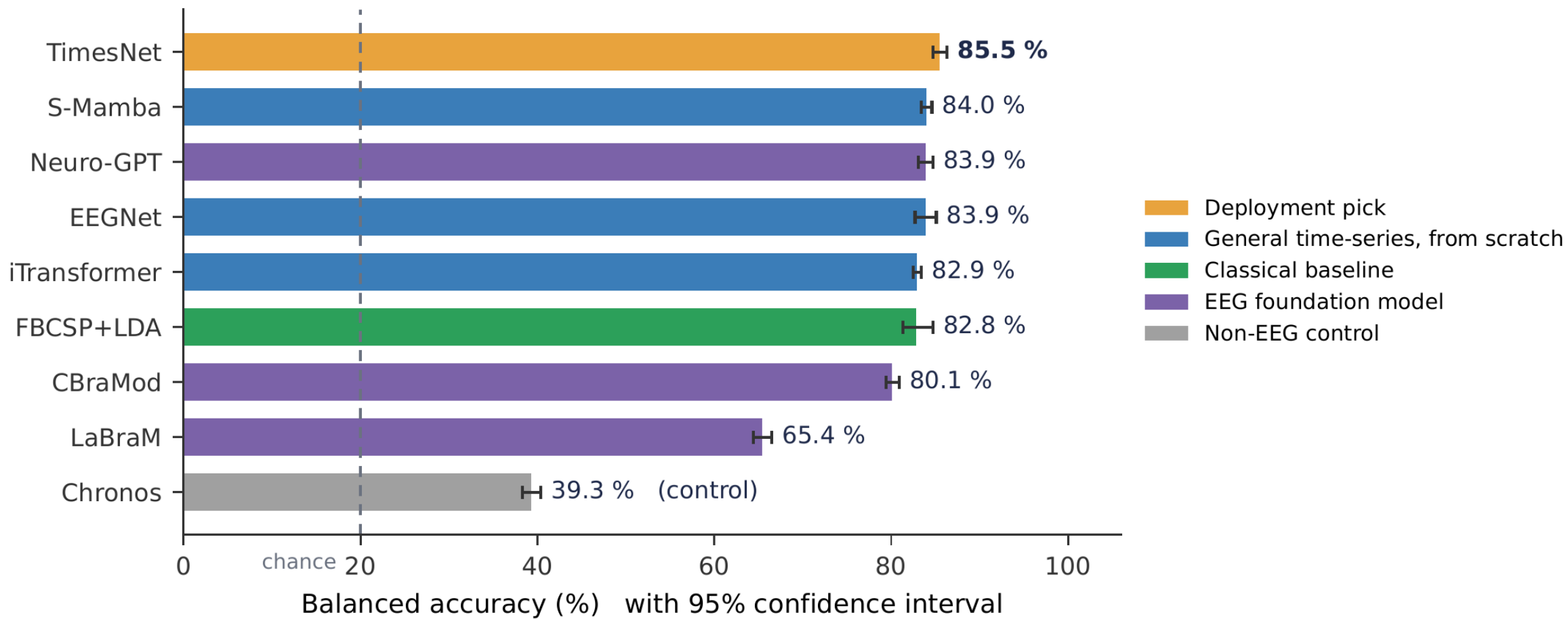
Native preprocessing and three adaptation regimes



- LaBraM at one third of its pretrained weights
- LoRA on the classifier, never the encoder

No champion in this benchmark

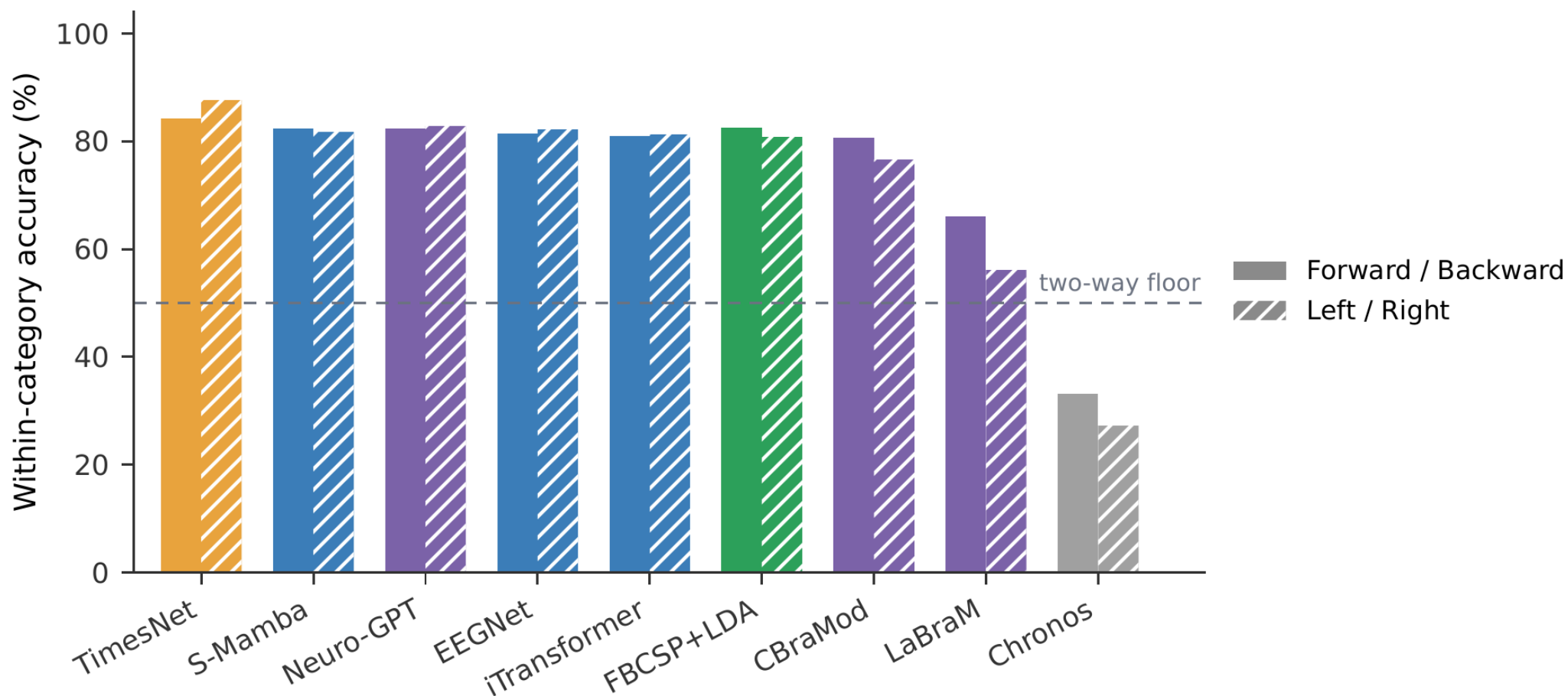
Protocol B: cross-recording, the deployment axis



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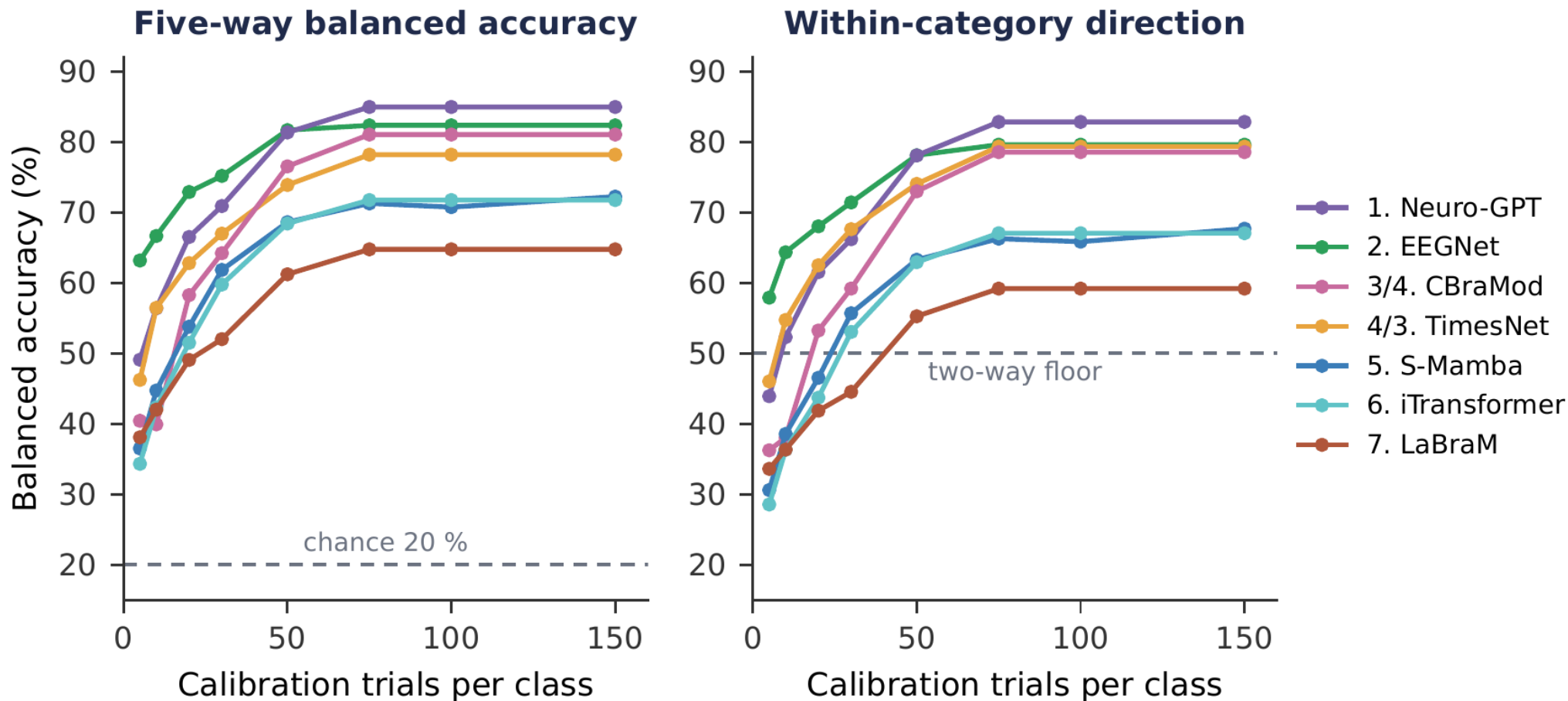
RQ0 Direction decoded, not motion only

Direction is resolved within movement category (Protocol B)



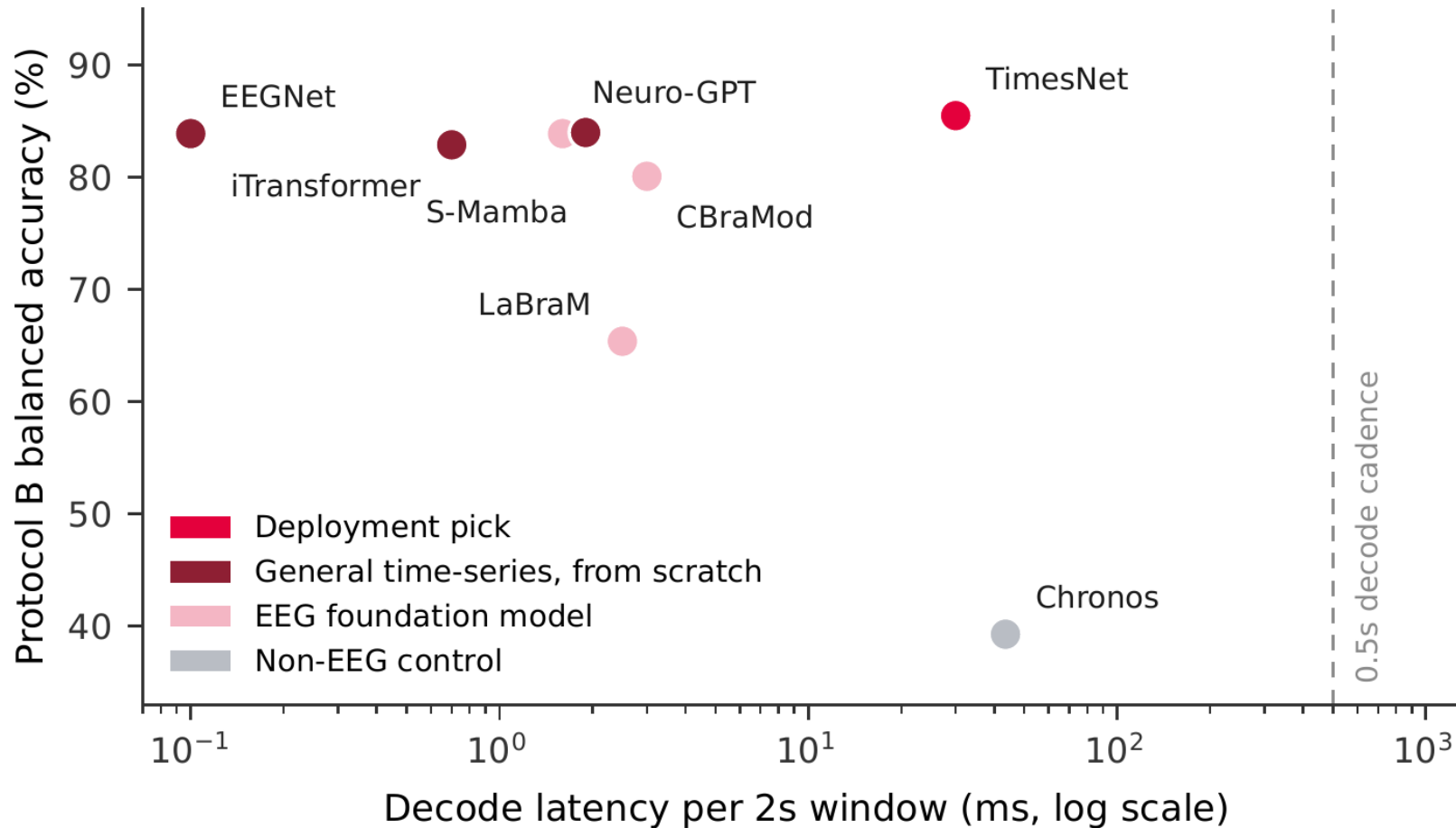
RQ3 Scarce data changes the winner

Sample efficiency in the low-data calibration regime



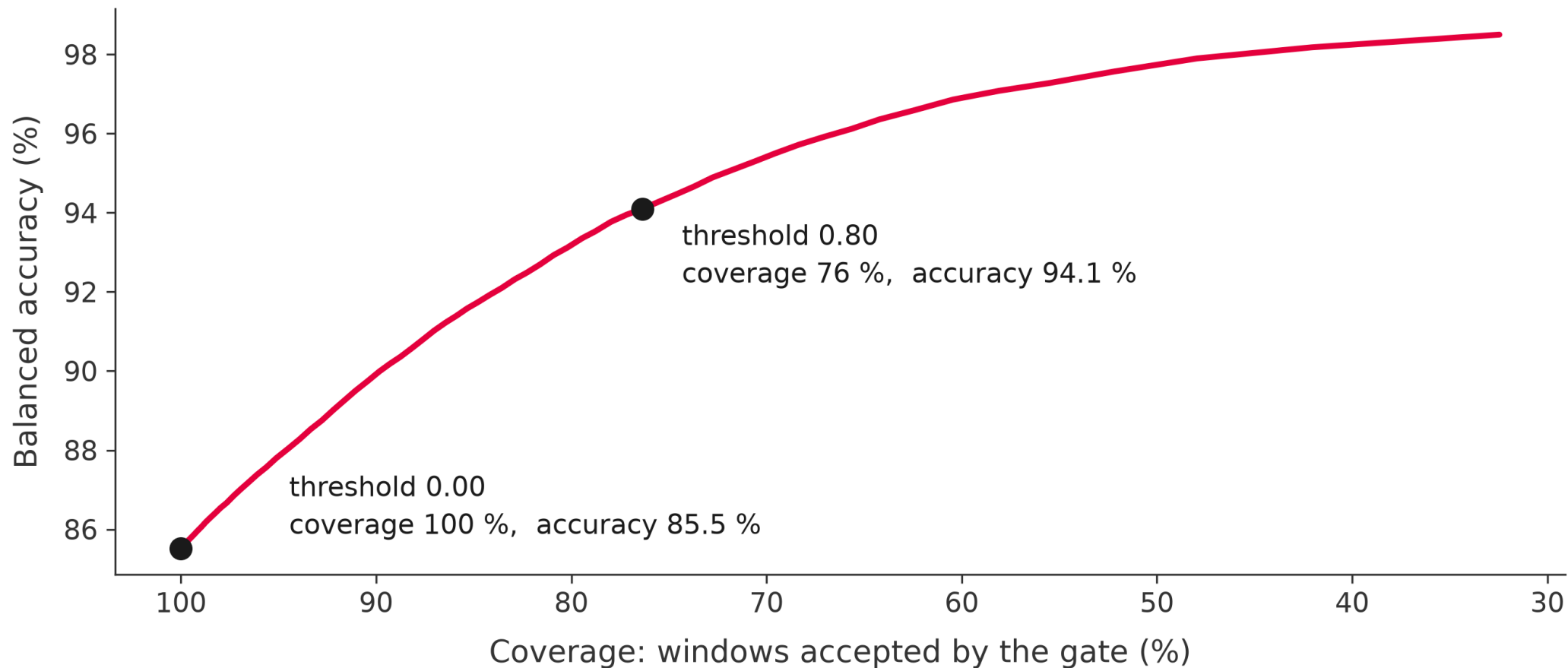
u^b Every decoder latency

Every decoder clears the real-time bound with wide margin



RQ4 TimesNet behind a gate

Confidence gate: rejecting low-confidence windows buys accuracy



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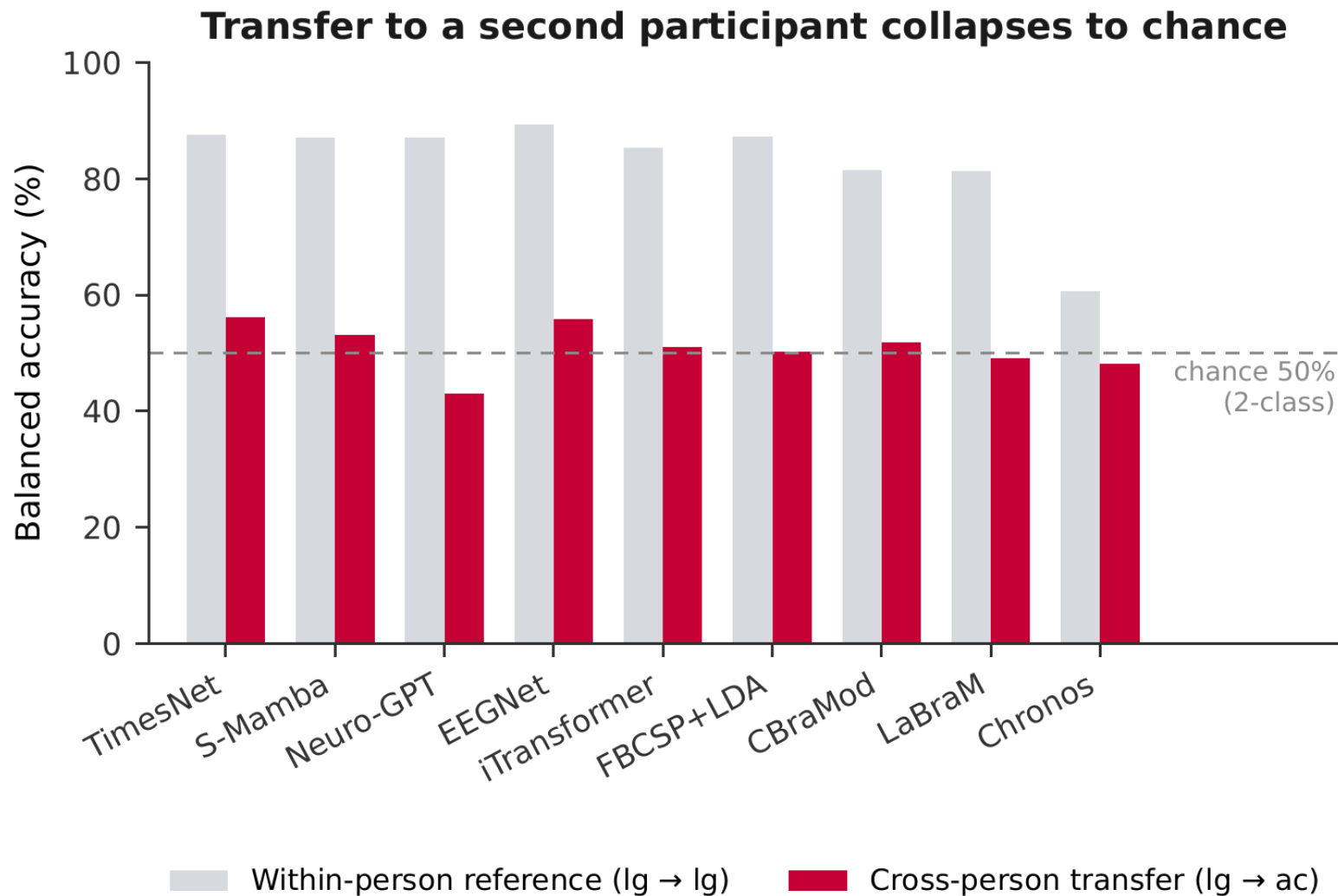
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Protocol C: it does not transfer yet



Deployment tests

Uncalibrated, on the wheeled surrogate

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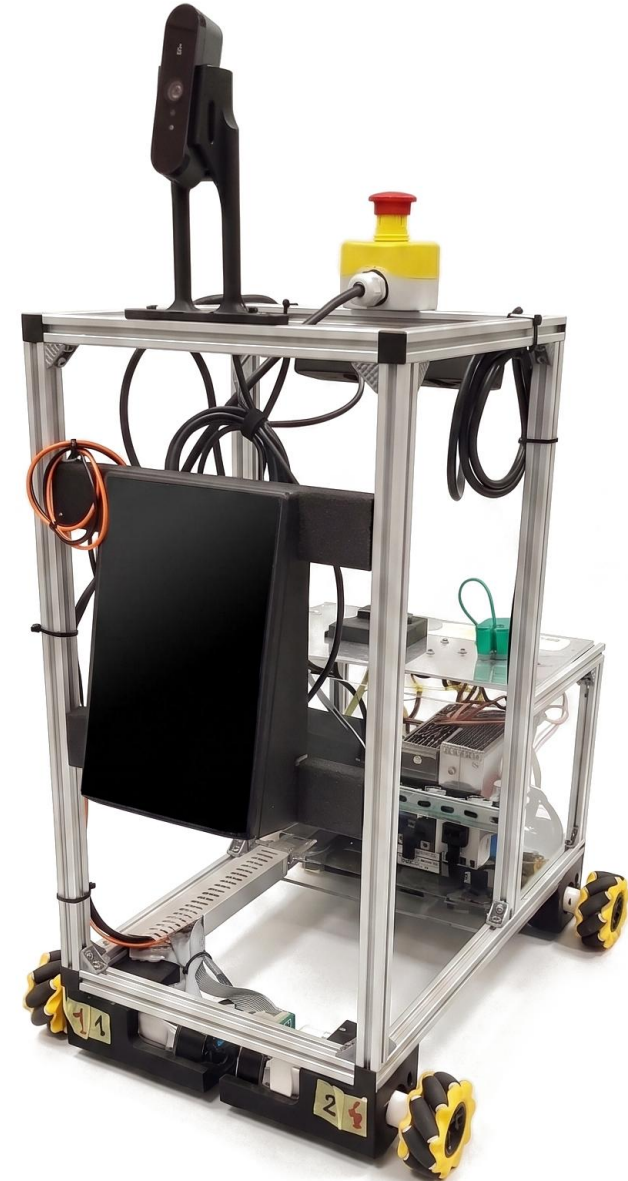
Conclusion

For an uncalibrated, unseen participant

- Protocol C predicted near-chance transfer
- Live behaviour exceeded expectation

Unrecorded: observation, not evidence

- EEGNet failed, is accuracy better?



Deployment live demo

Unrecorded session, qualitative observation

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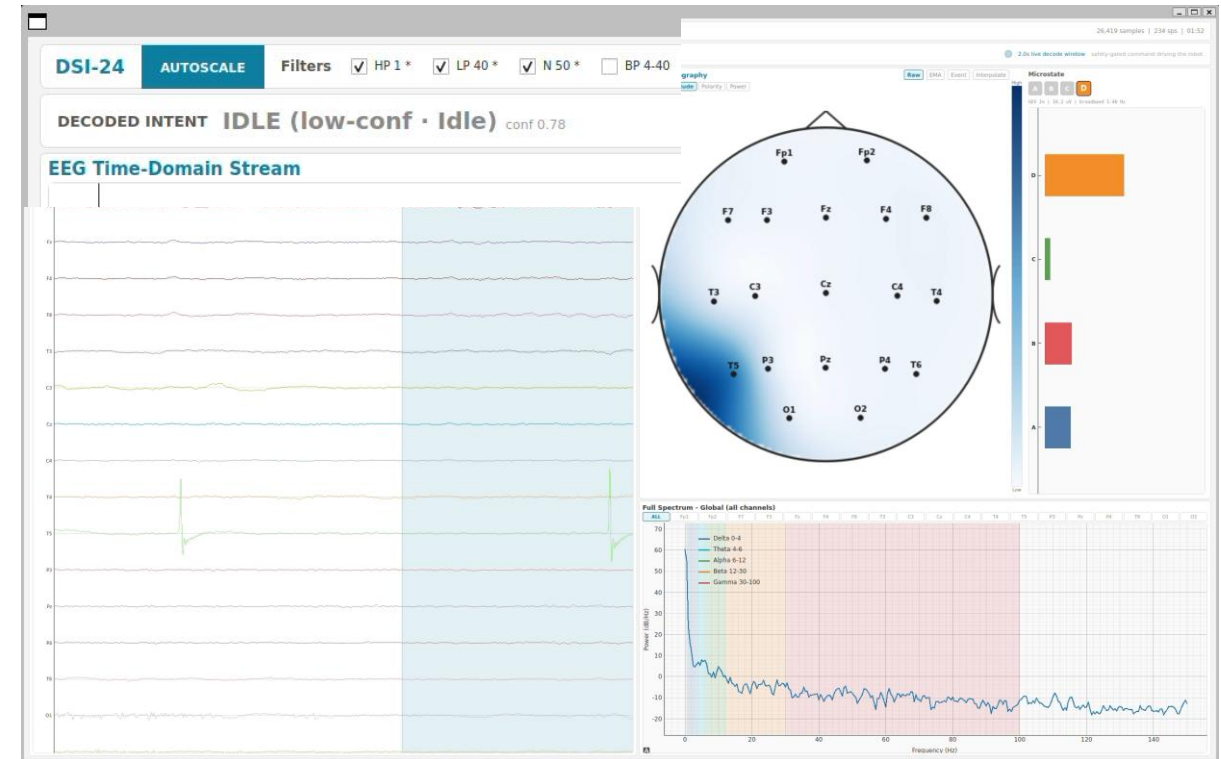
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NeuroDrive self-test

- Forward, Left, Right, **not** Backward
- Gate at 0.80, machine in simulation
- Artefact rejection off (no forced idle)

`TimesNet_DEPLOY_all-lg-data.pth`



u^b

Conclusion

Deployment tests, accuracy is priority, live demo

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Five questions, five answers

Is direction really decoded, or just movement?

RQ0 Yes.

Do EEG foundation models beat training from scratch?

RQ1 No. Joins the leading group, none leads it.

Does the pretraining corpus matter at all?

RQ2 Yes, but not because of pretraining hours or model size.

Who wins when calibration data is scarce?

RQ3 EEGNet. Resolves direction earliest.

Can the winner drive the robot safely, live?

RQ4 Yes, gated and calibrated per patient.

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u^b Limits of the experiment

Recording

- More participants, on platform, impaired, and imagined gait
- Compound directions spanning a change of movement
- Eye movement recorded alongside EEG

System

- Per patient calibration and fine-tuning
- Artefact rejection tuned and confidence scaled per decoder
- Shorter windows to reduce delay from intent to command

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u^b Gaps in the benchmark

Fair comparison

- Full protocol B rerun on a shared harness and budget
- Project every backbone output to a common head width
- Correct LaBraM weights and align LoRA modules

Stronger evidence

- Refit FBCSP+LDA without per epoch scaling
- Add a Riemannian baseline for covariance matrix of windows
- Classification head dropout and augmentation ablation study

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- 5-class dry-electrode in-motion dataset
- 9 decoders, 1 harness, 3 protocols, 1 classifier, 18 augmentations
- Decoders inseparable at the top (TimesNet highest)
- Cross-person transfer near chance
- Software: terminal UI, neuro-drive, recorder
- Self-test: exercised live, not measured
- Corrections and improvements to apply in next run

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Thank you for your attention

Questions?

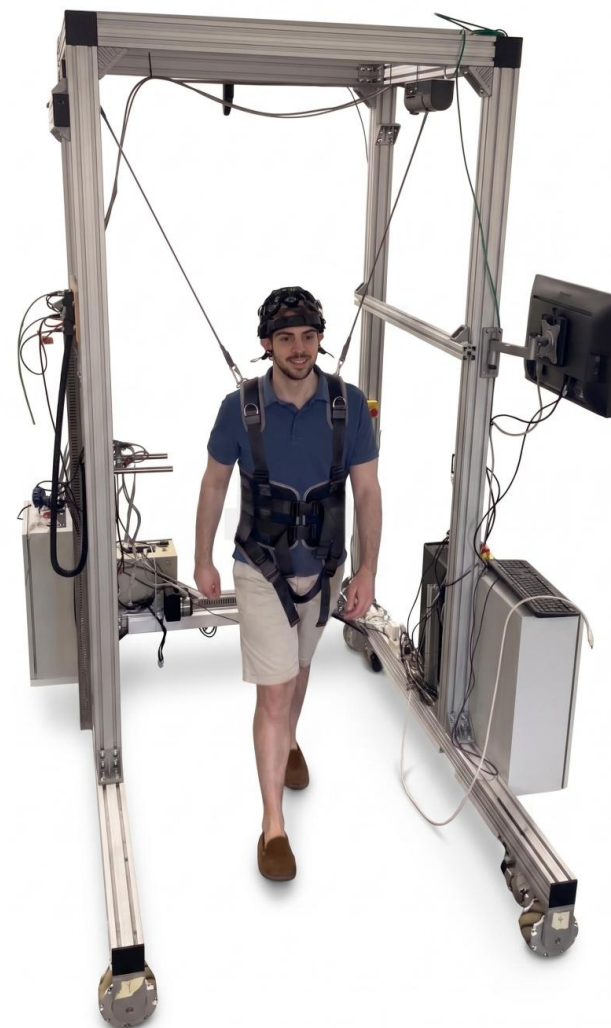
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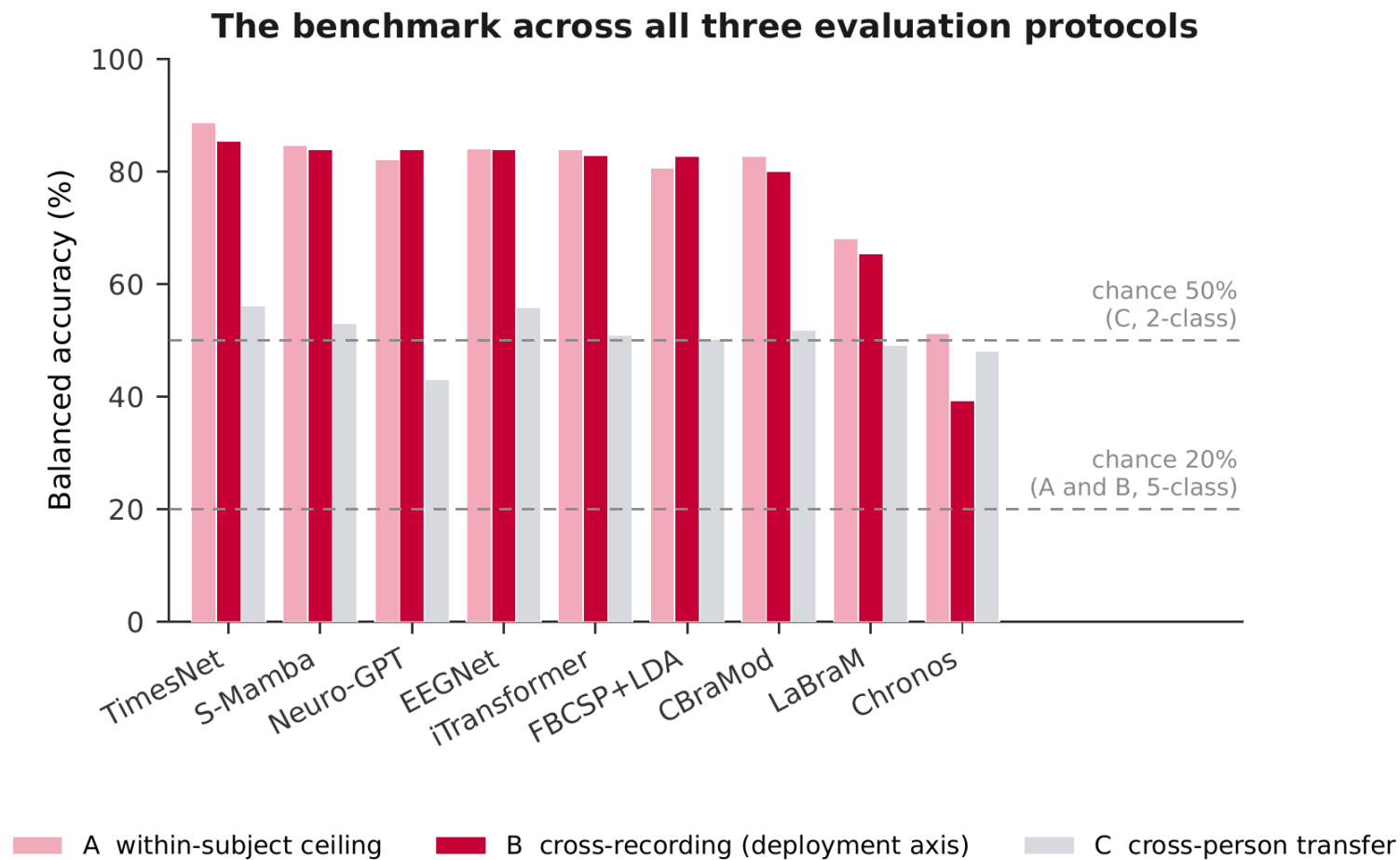
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Backup slides

B

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Backup: all three protocols



Neuro-GPT and FBCSP score higher on the harder Protocol B than on A. The ceiling is not leakage.

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Backup: conditioning decisions

Applied

- High-pass 1Hz against dry-electrode drift, above the MRCP band on purpose
- Low-pass 40Hz, notch 50Hz with Q of 30
- Per-epoch z-score for all nine decoders

Omitted, each by argument

- ICA: sources move against the montage during gait
- Re-referencing: electrode positions only approximate
- Epoch rejection: too little data to discard any

u^b Backup: the safety layer

Four gates before a command

- Artefact screen $\rightarrow$ confidence $\geq 0.80 \rightarrow$ rolling fail-safe $\rightarrow$ hysteresis

Asymmetric by design

- Three agreeing windows to start, two to stop, one frame on hard failure

Every failure path resolves to Idle

- Two independent 0.4s watchdogs, buffer warning after 5 s of silence

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Backup: the harness

Held identical for all nine

- Same 500 recordings, same windows, split seed 42
- Five seeds, batch 32, up to 80 epochs, patience 15
- One linear head, cross-entropy, balanced accuracy
- 120 trials at 20 epochs, judged on validation only
- No architecture modified

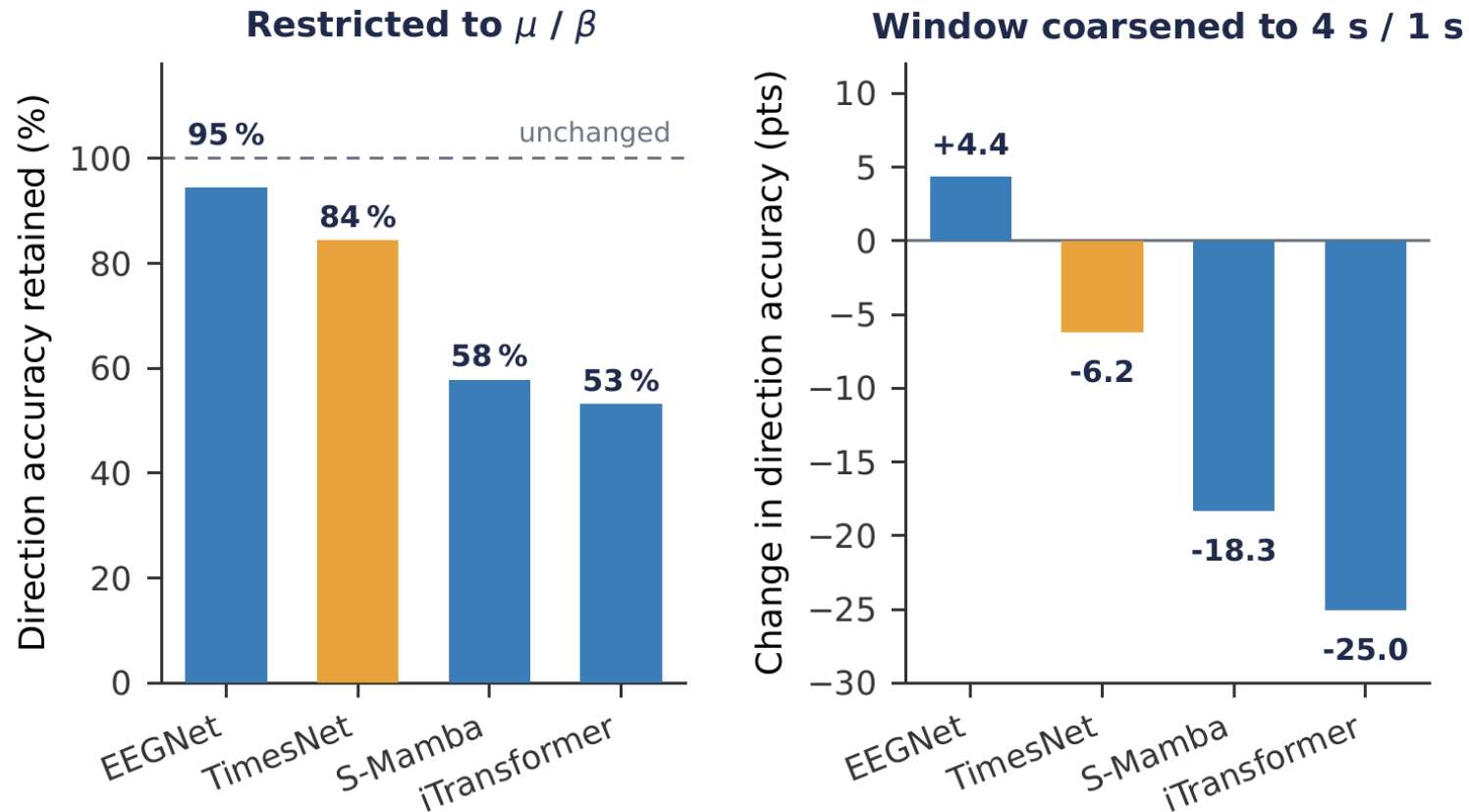
Tuned per model

- Learning rate, weight decay, augmentation
- Class-imbalance handling, mixup and cutmix

u^b

Backup

The same split under two unrelated manipulations



Two reduced-budget controls, run separately and for different reasons. Compare within each panel only, never against the headline table. The three foundation models are omitted: they sit below the two-way floor at both settings of both studies, so any ratio or difference computed on them would report noise.

Backup: Limitations

One training participant bounds every claim

One fixed partition, seeds vary weights, not folds

Head capacity follows backbone width, not equalised

LaBraM's checkpoint was two thirds untrained

Live sessions unrecorded, observation not evidence

Gate reads raw softmax, never calibrated

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Backup: Protocol B crown run

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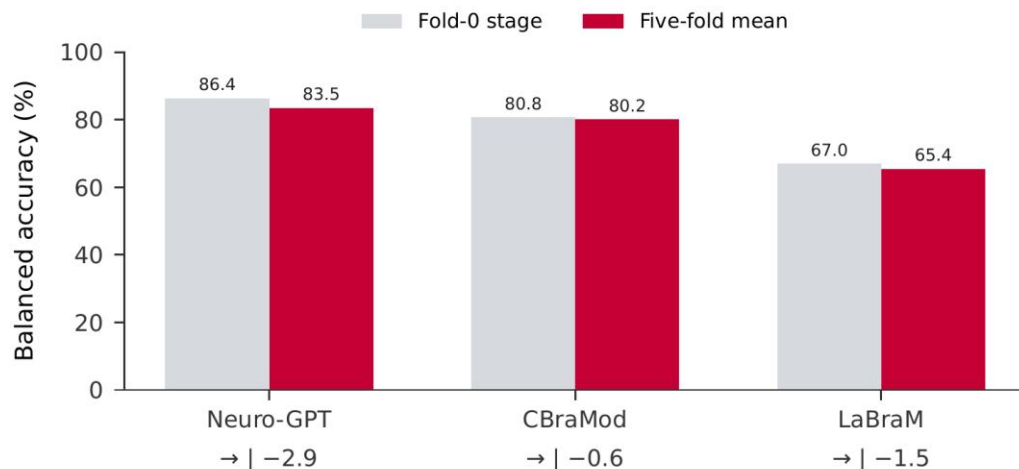
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Fold-0 stage against the five-fold mean



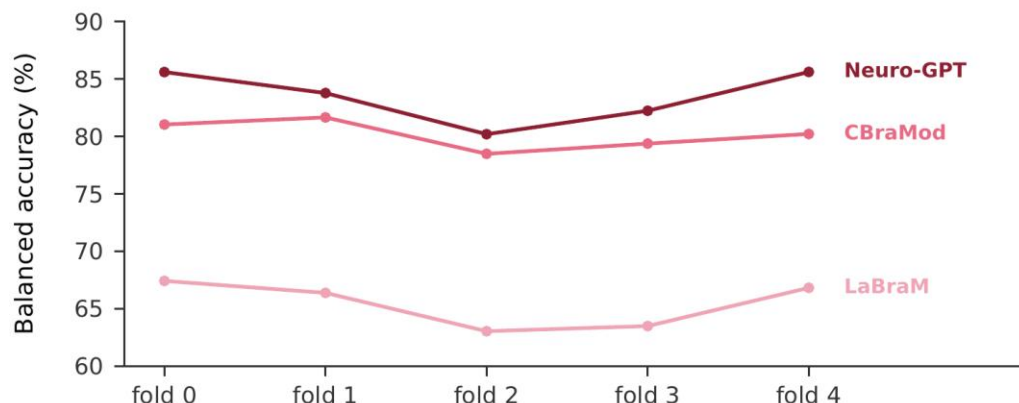
One fold decides nothing

- Fold 0 flatters all three fine-tuned models, by up to 2.9 points
- Folds 2 and 3 are hardest for every model: the split, not noise

From full benchmark to one question

- Can any challenger beat the crown for deployment?
- Five candidates re-searched, best of two configurations

Per-fold means within the five-fold run



Compute follows clinical promise

- Neuro-GPT carried the pretrained case, strongest of three
- CBraMod, LaBraM, Chronos: lower bounds by design

Defensible now, retrained later

- A future iteration reruns all nine under one budget

Backup: Why LaBraM underperforms

LaBraM as benchmarked: one third of the checkpoint was pretrained

- 5.8 million parameters used by the model, compared one by one with the authors' original checkpoint



■ pretrained, transferred from the original

33%

1.94M parameters

▨ started from scratch in the file benchmarked

67%

3.87M parameters

— every feed-forward layer, all 12 blocks	3.85M
— both positional tables	15k
— final normalisation	400

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Backup

DSI-24 **AUTOSCALE** **Filters** ☒ HP 1 * ☒ LP 40 * ☒ N 50 * ☐ BP 4-40

DECODED INTENT **IDLE (low-conf · Forward)** conf 0.63

DSI-24 **AUTOSCALE** **Filters** ☒ HP 1 * ☒ LP 40 * ☒ N 50 * ☐ BP 4-40

DECODED INTENT **IDLE (holding · Forward)** conf 0.87

DSI-24 **AUTOSCALE** **Filters** ☒ HP 1 * ☒ LP 40 * ☒ N 50 * ☐ BP 4-40

DECODED INTENT **▶ FORWARD** conf 0.88

```
(.venv) ~/master_thesis/project$
```

```
python src/streamer.py --neuro-drive --no-artifact-gate --model TimesNet_DEPLOY_all-1g-data.pth
```

Backup: The switch to TimesNet

The benchmark pick became the deployed model

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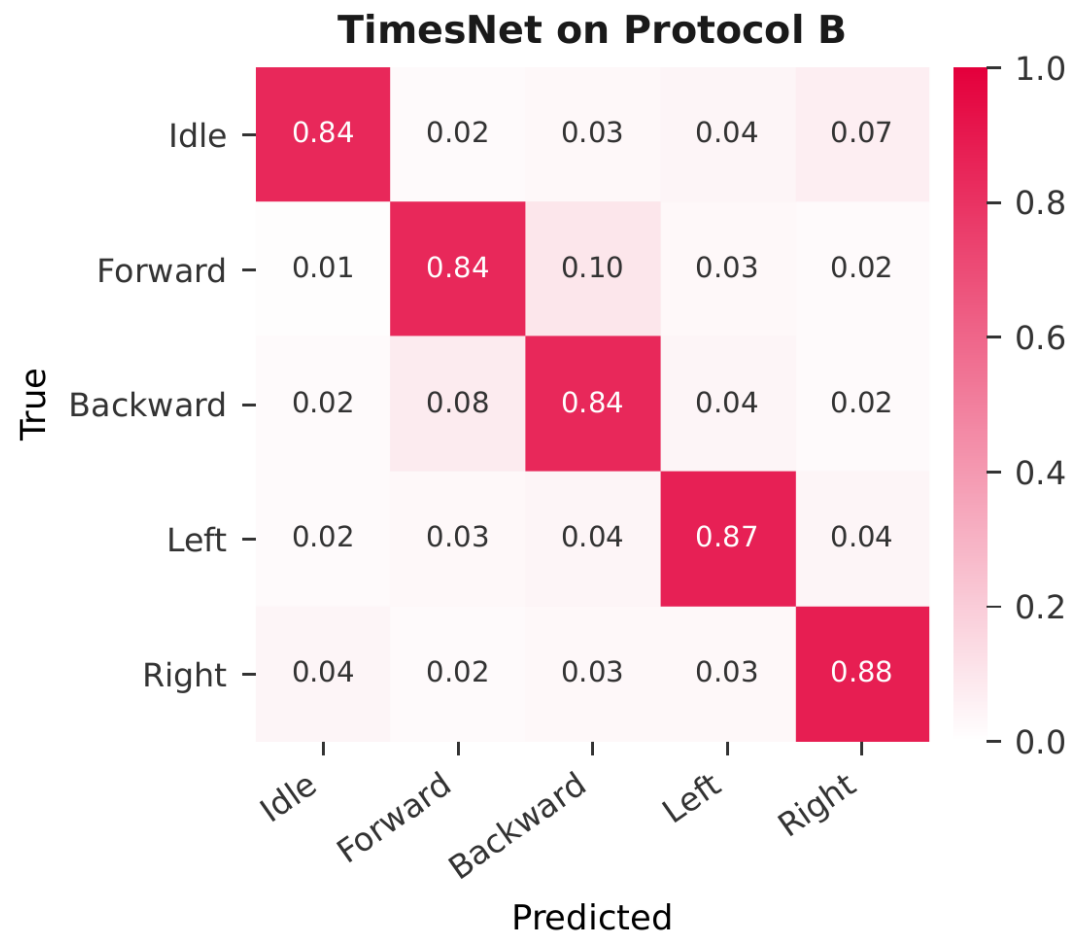
EEGNet failed, is accuracy better?

Recall at least 0.84 on every class

- Not driven by the easy Idle class

Errors remain within movement

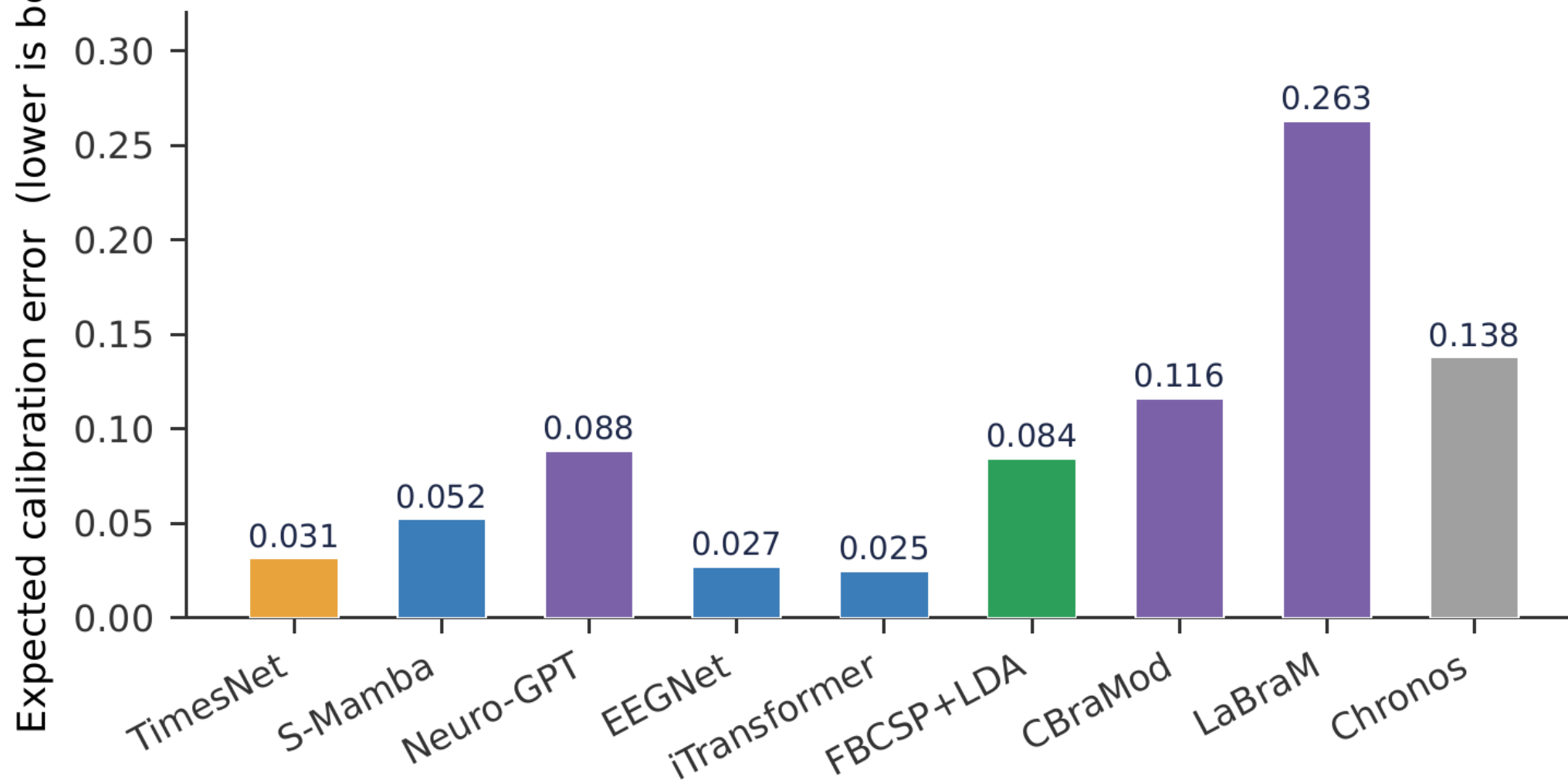
- Forward-Backward confusion



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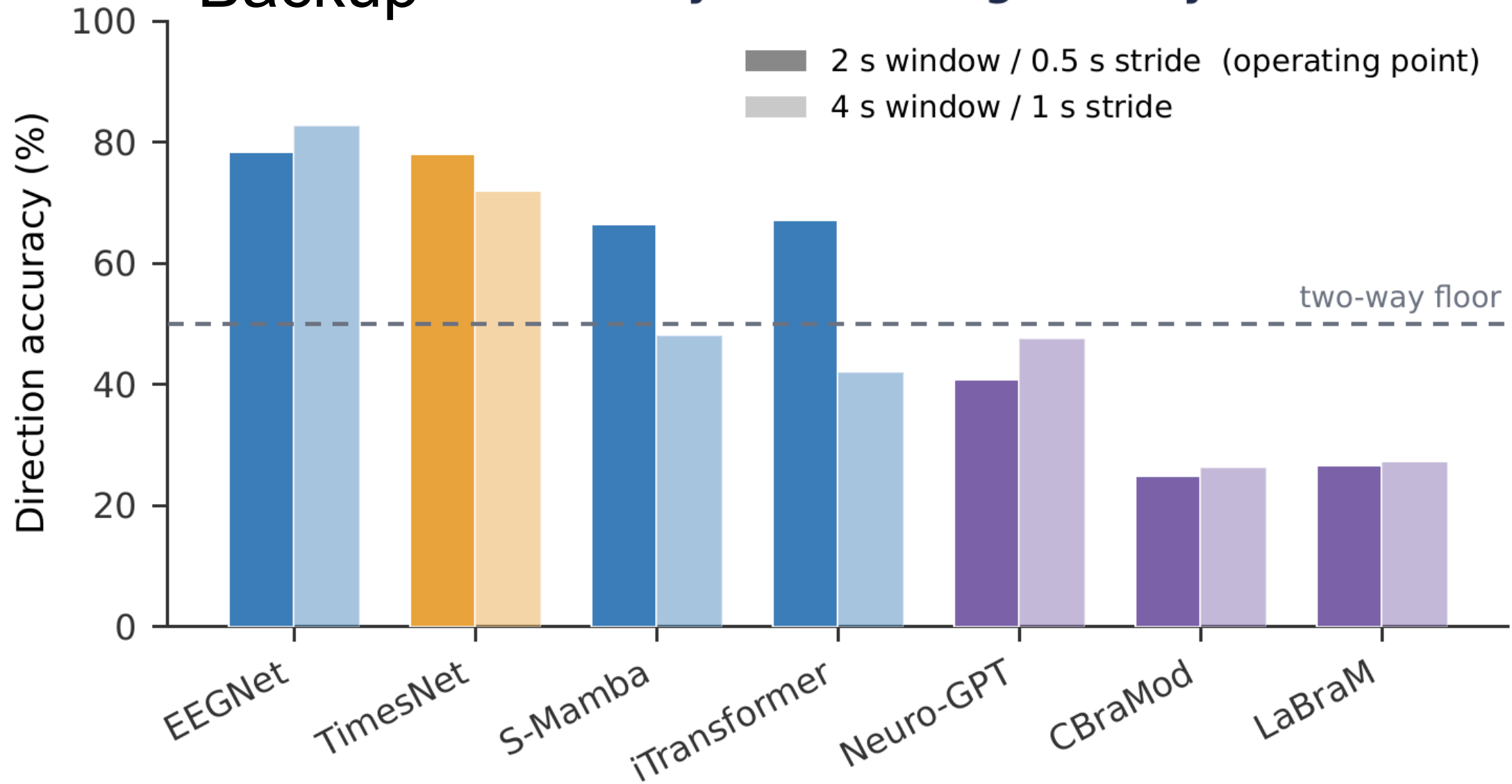
Backup

Calibration on the deployment axis

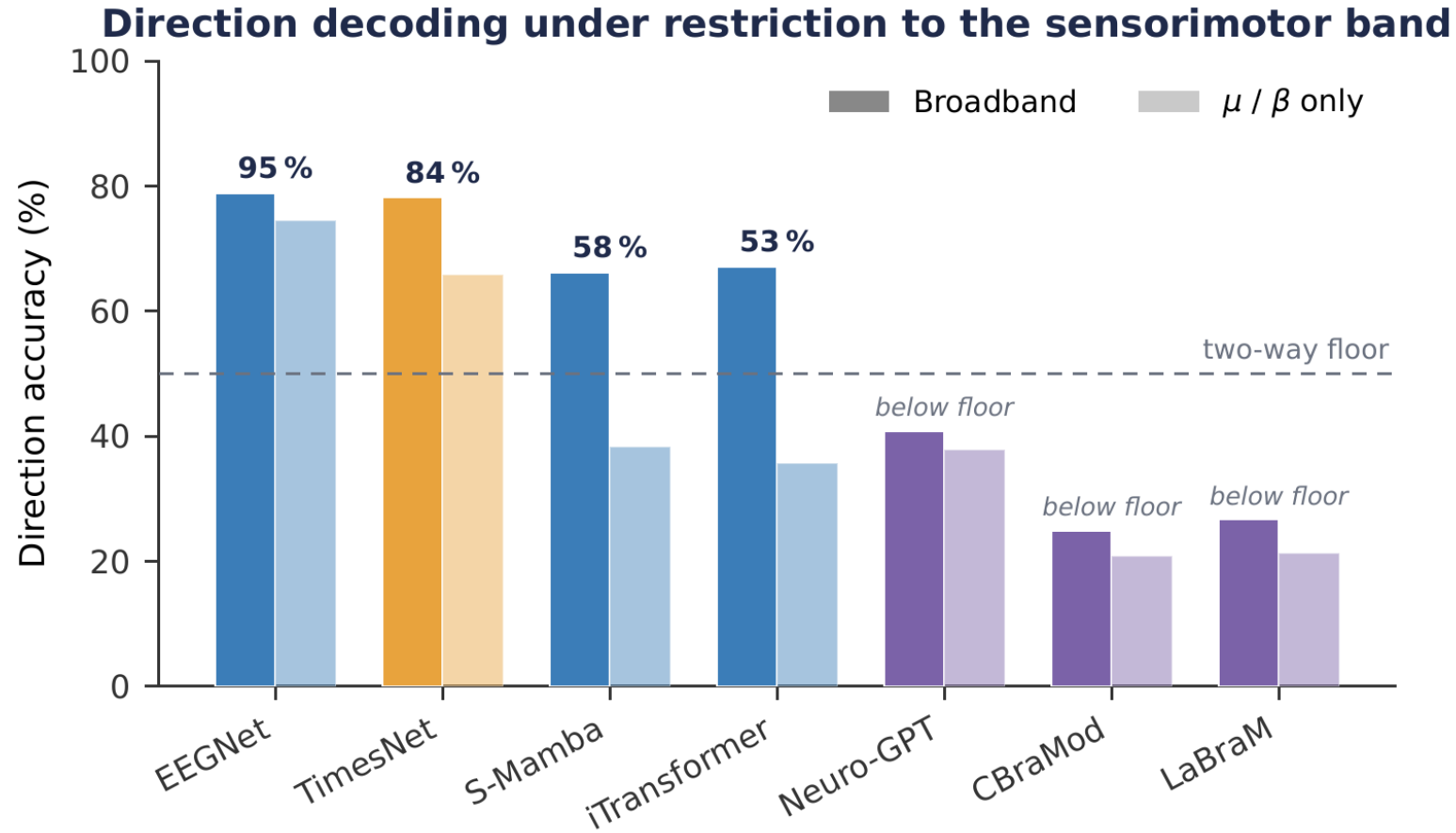


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Backup Sensitivity to window geometry



Backup: Band-restriction Control



Bold percentages give the fraction of broadband direction accuracy retained under band restriction, shown only where the broadband value clears the two-way floor; below it the model is not resolving direction at all. Reduced-budget control: compare within this figure only, never against the headline table.